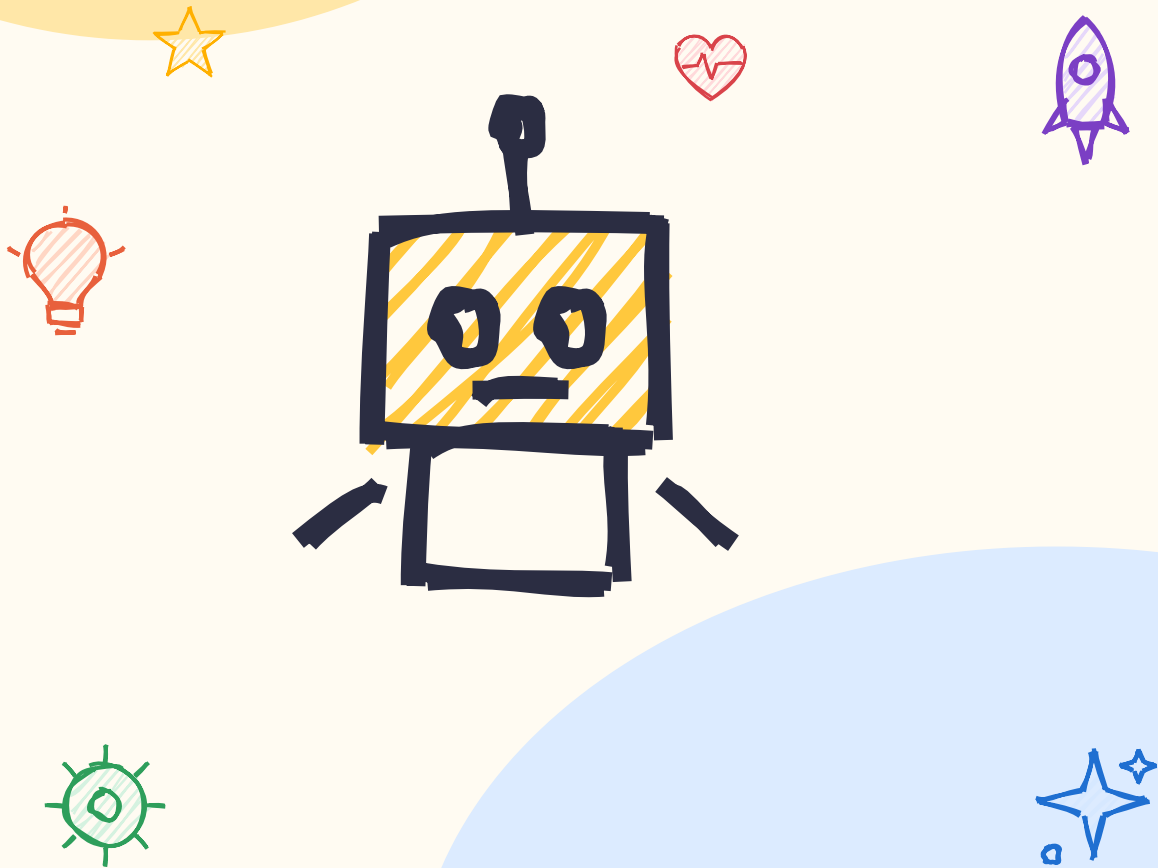


Pak-Turk Maarif presents

Robotics & AI Club

Learning Pathways 2026-27

A hands-on journey into electronics, coding, robotics and artificial intelligence for ages 5 to 17.



CURRICULUM DEVELOPED BY

Qasim Mushtaq

MS Robotics and AI

Little Explorers
Ages 5-7

Junior Makers
Ages 8-11

Young Innovators
Ages 12-14

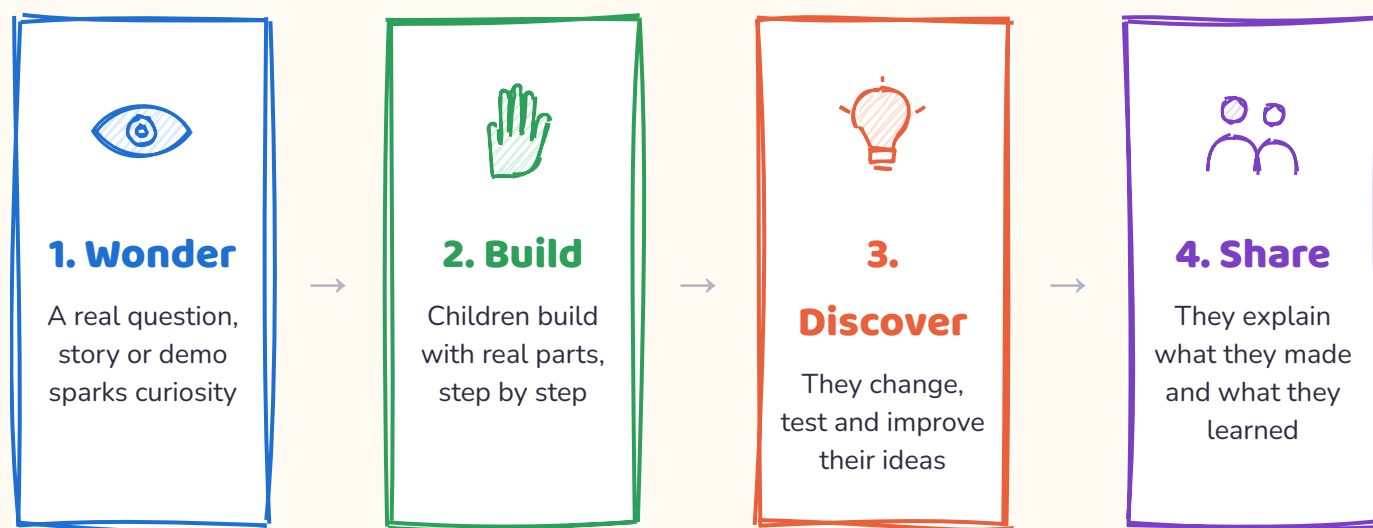
Tech Leaders
Ages 15-17

Dear parents,

Welcome to the Robotics & AI Club

Technology is shaping the world our children will grow up in. Our club helps them become **creators, not just users**, of that technology. Over 16 classes (two a week for 8 weeks), students in every age group build real projects, write real code and learn to solve problems as a team. Teams can then design their own project for an end-of-batch showcase.

What happens in a class?



The programme at a glance



Learn by doing

Every class ends with something your child built themselves.



Simulate, then build

Children test ideas safely on screen before using real parts.

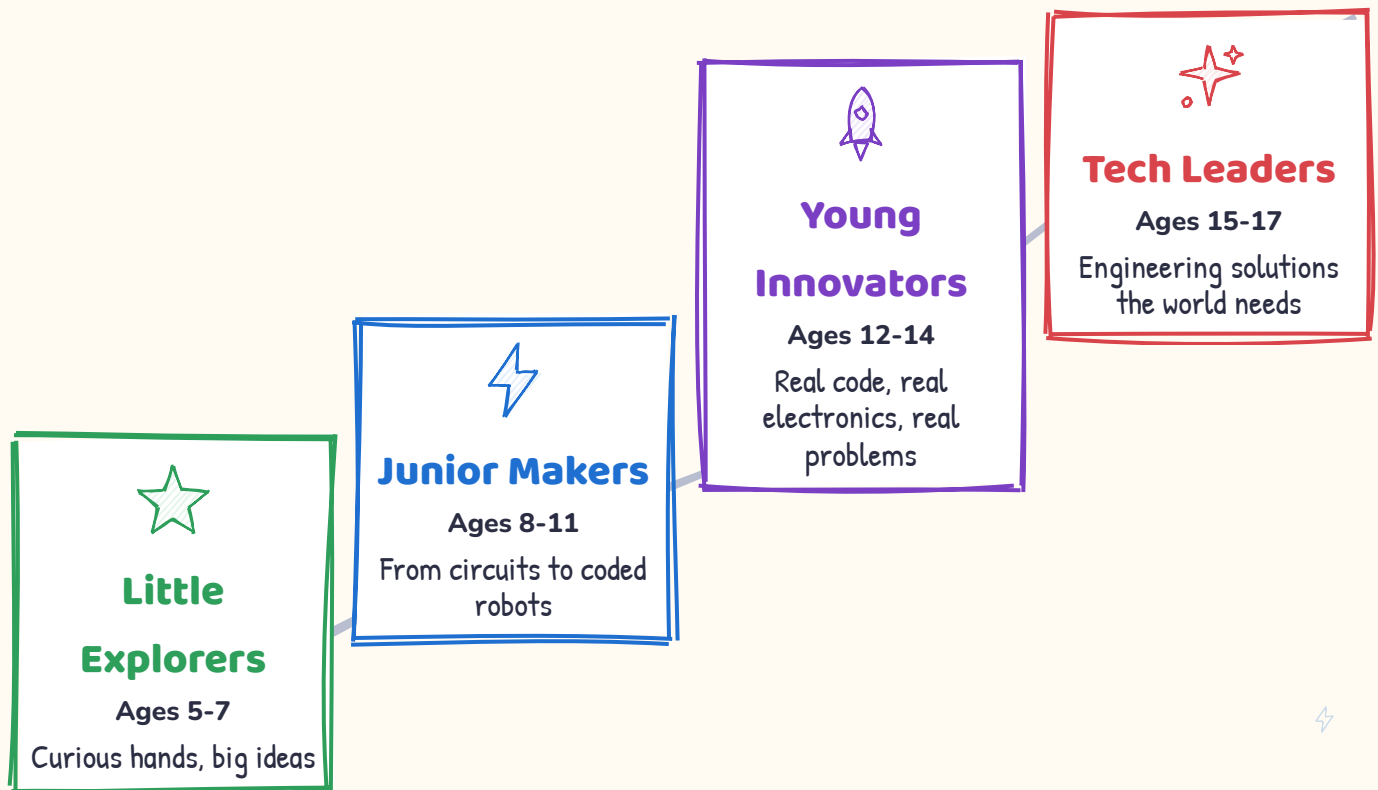


Safe and responsible

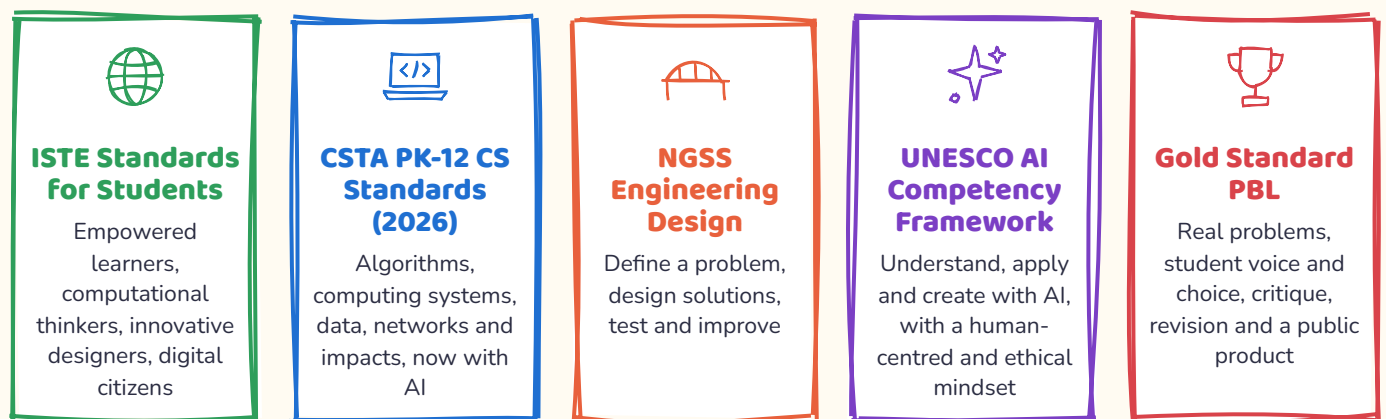
Age-appropriate tools, supervised equipment and a focus on ethics.

Four pathways, one journey

Each age group follows its own 16-class pathway: two classes a week for 8 weeks. The skills grow step by step, so a child who joins at 5 can keep building all the way to 17.



Built on international standards



ISTE: International Society for Technology in Education · CSTA: Computer Science Teachers Association · NGSS: Next Generation Science Standards · PBL: project-based learning

Linked to the UN Global Goals



Every activity carries badges for the UN Sustainable Development Goals it connects to, from clean energy to good health.

Ages 5-7

Little Explorers

Curious hands, big ideas



Young children learn best with their hands. In Little Explorers, every class is a playful mini-adventure: your child lights up paper circuits, powers an LED with lemons, programs a real Arduino board with colourful picture blocks, makes fruit play music with Makey Makey and builds their first moving robots. The teacher prepares all the wiring, so children can focus on pressing, testing, creating and explaining.

The learning journey



Classes 1-4

Circuits & Power

- Discovers that electricity needs a complete loop
- Builds a light-up card and a working doorbell
- Programs a traffic light with picture blocks



Classes 5-8

Sensing the World

- Uses sensors that notice light, heat, water and movement
- Makes a lamp, a plant alarm and a room alarm work on their own
- Understands if-then decisions



Classes 9-12

Move & Communicate

- Makes robot arms wave and railway gates move
- Improves a vibrating robot after testing it
- Sends secret messages in Morse code



Classes 13-16

Code, Play & AI

- Turns fruit into a piano with Makey Makey
- Creates an animated story and a game with a homemade controller
- Learns how machines sort information using simple rules



After class 16 • Project showcase!

Teams design and build their own project, then present it at the showcase.



By the end of 16 classes, your child can...



Build a simple circuit that lights an LED



Program an Arduino with picture blocks



Explain how a sensor helps a machine decide



Make a motor move a robot arm or gate



Create an animated story and a simple game



Make a prediction, test it and improve it



Explain in simple words how computers sort information



Share their creation proudly with others

Skills they grow

Curiosity



Creativity



Problem solving



Coding



Electronics



Teamwork



More dots = more focus in this pathway

Tools they will use

Arduino (picture blocks)

mBlock / ArduBlock

Makey Makey

ScratchJr

Scratch

PhET simulations

Free, education-friendly tools, used in class with teacher accounts where needed.



Grand finale: project-based learning

A small invention such as a light-up bedroom gadget, a smart plant helper or a musical instrument, shown to parents at the showcase. Children practise **planning, teamwork, giving feedback, improving their work and presenting** to an audience.

Try this at home

Ask your child to 'program' you: they give you step-by-step instructions to make a sandwich or walk to the door, and you follow them exactly.



Junior Makers move from 'how does it work?' to 'I built it!'. Your child wires real circuits, programs Arduino boards with blocks that reveal real code, and combines several sensors, motors and screens into smart gadgets, from a sun-tracking solar panel to a line-following robot. Every idea is tested safely in a simulator before it is built.

The learning journey



Classes 1–4

Circuits, Code & Sensing

- Wires and measures circuits with a multimeter
- Builds a reaction-timer game and a weather station
- Combines two sensors in a smart dustbin



Classes 5–8

Smart Home & Energy

- Protects a home from rain, fire and gas
- Controls motor speed with temperature
- Builds a solar panel that follows the sun



Classes 9–12

Security, Design & Robots

- Programs a burglar alarm with a secret code
- Designs a part in 3D for printing
- Builds a line-following robot and a phone-controlled lamp



Classes 13–16

Health, Web, Games & AI

- Makes a step counter with a motion sensor
- Writes a first webpage and a game with a real controller
- Trains an AI and checks it for fairness



After class 16 • Project showcase!

Teams design and build their own project, then present it at the showcase.



By the end of 16 classes, your child can...



Wire circuits on a breadboard safely



Program an Arduino using block coding



Combine 2-3 sensors with motors and screens



Measure voltage with a multimeter



Use loops, conditions and variables



Design a 3D part and build a robot



Create a webpage and a game controller



Train and test a simple AI model

Skills they grow

Curiosity



Creativity



Problem solving



Coding



Electronics



Teamwork



More dots = more focus in this pathway

Tools they will use

ArduinoBlocks

mBlock 5

Tinkercad Circuits

Tinkercad 3D

MIT App Inventor

Teachable Machine

HTML & CSS

Free, education-friendly tools, used in class with teacher accounts where needed.



Grand finale: project-based learning

A working gadget such as an automatic pet feeder or a smart study lamp, planned and built by your child's team. Children practise **planning, teamwork, giving feedback, improving their work and presenting** to an audience.

Try this at home

Go on a 'sensor hunt' at home: find everything that senses light, sound, motion or temperature (automatic doors, phones, fridges) and talk about how it works.

Young Innovators

Real code, real electronics, real problems



Young Innovators step into real engineering. Your child writes real code in C++ and Python, builds robots that scan their surroundings, creates phone apps that control hardware, sends radio messages between devices and builds a Wi-Fi plant-watering system with its own app and 3D-printed case.

The learning journey



Classes 1-4

Electronics & Mechanisms

- Builds a timer circuit without any code
- Monitors heart rate and saves classroom energy
- Controls precise stepper motors in a plotter



Classes 5-8

Robots & Security

- Programs a robot that scans and avoids obstacles
- Builds a Bluetooth car with their own phone app
- Creates a gesture-controlled arm and an RFID lock



Classes 9-12

Power & Wireless

- Builds a safe rechargeable power pack
- Links two devices by radio
- Creates a Wi-Fi plant system and its mobile app



Classes 13-16

Software, AI & Design

- Writes Python games and a portfolio website
- Builds a chatbot and learns the limits of AI
- Designs a 3D-printed product enclosure



After class 16 • Project showcase!

Teams design and build their own project, then present it at the showcase.



By the end of 16 classes, your child can...



Write and debug Arduino code in C++



Write Python programs with functions



Build Android apps that control hardware



Connect devices by Bluetooth, radio and Wi-Fi



Explain timers, relays and battery safety



Design and prepare a 3D print



Build a website to show their work



Use AI tools critically and responsibly

Skills they grow

Curiosity



Creativity



Problem solving



Coding



Electronics



Teamwork



More dots = more focus in this pathway

Tools they will use

Arduino IDE

ESP32

Wokwi

Falstad simulator

MIT App Inventor

Python

HTML, CSS & JavaScript

Tinkercad & BlocksCAD

Free, education-friendly tools, used in class with teacher accounts where needed.



Grand finale: project-based learning

A team project such as an earthquake alarm or a Wi-Fi mailbox notifier, presented with a short pitch. Children practise **planning, teamwork, giving feedback, improving their work and presenting** to an audience.

Try this at home

Ask your child to explain one gadget at home (a TV remote, a smartwatch, a Wi-Fi router) using the words **input, process and output**.



Tech Leaders work like real engineers. Your child designs solar-powered and wireless systems, builds autonomous robots with control algorithms, analyses real data with Python, trains machine learning models, secures internet-connected devices and designs products for 3D printing, then presents solutions to real problems.

The learning journey



Classes 1-4

Power, Autonomy & Health

- Designs an off-grid solar device with an energy budget
- Tunes a PID-controlled maze robot
- Builds a fall-detection wearable for older people



Classes 5-8

Connected Systems

- Builds wireless sensor networks and a cloud smart home
- Creates a live smart-parking dashboard
- Designs a two-factor security locker



Classes 9-12

Data, AI & Web

- Collects and analyses real data in Python
- Trains and evaluates machine learning models
- Builds web APIs that control devices



Classes 13-16

Security, Robots & Product Design

- Finds and fixes security weaknesses ethically
- Builds a fire-fighting robot
- Designs parametric 3D parts and a motion-controlled game



After class 16 • Project showcase!

Teams design and build their own project, then present it at the showcase.



By the end of 16 classes, your child can...



Build IoT systems with Wi-Fi and cloud dashboards



Program feedback control (PID) for robots



Analyse real data with Python



Train and evaluate a machine learning model



Apply basic cybersecurity to real devices



Design low-power and solar systems



Design parametric 3D parts



Present a technical solution to an audience

Skills they grow

Curiosity



Creativity



Problem solving



Coding



Electronics



Teamwork



More dots = more focus in this pathway

Tools they will use

Arduino IDE

ESP32

Arduino Cloud

Python (pandas, scikit-learn)

Pygame & Flask

BlocksCAD & Cura

picoCTF

Free, education-friendly tools, used in class with teacher accounts where needed.



Grand finale: project-based learning

An end-to-end system such as an autonomous delivery robot or an IoT product, with a real user, a prototype and a pitch. Children practise **planning, teamwork, giving feedback, improving their work and presenting** to an audience.

Try this at home

Ask your teen to pick one local problem (water, traffic, energy) and describe a tech solution they could build. Then ask: who would use it, and what could go wrong?

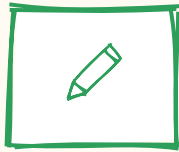
Learning by building real projects

Every pathway ends with two project classes. Our approach follows the internationally recognised **Gold Standard PBL** model by PBLWorks:



A real problem

Projects start with a question that matters



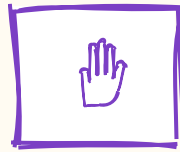
Research & plan

Children ask questions and find answers



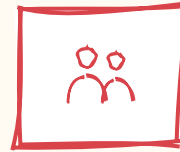
Real world

Real tools, real users, real impact



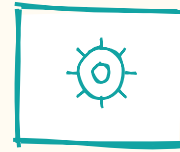
Voice & choice

Teams decide what and how to build



Feedback

Friends and teachers give helpful feedback



Improve

Build, test, fix, repeat



Show the world

A public showcase for families



Safe & responsible technology

- Low-voltage, child-safe electronics
- Hot glue, cutters and soldering handled by teachers for younger children
- AI tools used within their age rules; under-13s use AI only with the teacher
- Lessons on online safety, privacy and the ethics of AI



How you can help

- Ask "What did you build today?" and let them explain
- Celebrate mistakes: debugging is how engineers learn
- Save old boxes, caps and CDs for craft projects
- Come to the end-of-term project showcase



See you in the lab!

Robotics & AI Lab

Pak-Turk Maarif International Schools & Colleges · Islamabad Boys Campus, Chak Shahzad

For questions about the programme, please contact the Robotics & AI Club team through the school office.

Curriculum developed by Qasim Mushtaq, MS Robotics and AI